

Steering Is Not Personalization: Stable Behavioral Differences and the Adaptation Gap in AI Tutors

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Abstract

Prompting can make an AI tutor ask more questions, reveal fewer answers, or sound more supportive. It is tempting to call this personalization. We show that it is usually something narrower: a reliable shift in behavior that need not fit the learner. Using same-item responses from a large educational benchmark archive, we first ask whether models differ at all, which differences recur across tasks, and how a shared pedagogy prompt changes them. We then test whether the induced behavior agrees with human teacher actions, reflects learner-state diagnosis, or supports context-conditioned action selection. Models do exhibit recurring defaults. Help directness transports across teaching tasks and converges with independently inferred teacher actions; expressed confidence retains a stable ordering across four sources. Yet the same prompt moves every model sharply toward elicitation and away from direct help, with model-specific movement and residual model identity. This steering is not personalization. The prompt raises aggregate teacher-action agreement but suppresses direct instruction when human teachers chose telling. Models realize an assigned ASK or EXPLAIN action almost perfectly, while tested selectors remain near chance. Several attractive trait interpretations also fail: steerable actionability does not recur, relational communion largely restates warmth, and questionnaire agreeableness does not predict open behavior. AI-tutor evaluation should therefore treat behavior as a model–instruction–context interaction and separate default style, steering, action selection, and learning outcomes.

1 Introduction

People interacting with AI tutors quickly notice that systems teach differently. Given the same learner and problem, one model explains, another asks a question, and a third answers with unusual certainty. Before asking whether such differences are

stable, a more basic question is necessary: after the item and generation setup are controlled, do models exhibit systematic differences in educational behavior at all? If they do, a second question follows: are these differences properties that recur across teaching tasks, or artifacts of one benchmark, one prompt, or a generic authorship signature?

Most language-model personality research begins with human inventories or an assigned persona (Jiang et al., 2024; Lee et al., 2025). Those methods show that a model can produce trait-consistent answers, but they do not show that the same ordering appears when the model actually teaches. Inventory scores also change with wording and option order (Shu et al., 2024), and recent work questions whether human psychometric constructs retain their meaning for models (Han et al., 2025; Zierahn et al., 2026; Contreras, 2026).

Education makes the distinction consequential and testable. A tutor must decide whether to tell or elicit, whether to reveal an answer or preserve learner work, how much information to introduce, and how firmly to state a judgment. These choices can be compared across repeated teaching situations and against human teacher actions or learner-state labels. They also expose a distinction that is easy to miss in prompt engineering: reliably inducing a teaching move is not the same as selecting the right move for this learner now. A question can be useful after a productive error and obstructive when direct instruction is needed.

We therefore ask six questions. Do models differ in educational behavior under matched conditions? Which differences recur across teaching tasks, and do any extend beyond education? Are they distinct from length, formatting, and generic authorship cues? How much does a shared pedagogy prompt move them? Do models respond to that prompt in the same way? Finally, does a recurring or induced style predict context-appropriate teaching? We use “educational personality” only as shorthand for re-

curing observable tendencies in these settings, not for an inner state or a human-equivalent trait.

The study was made possible by an unusual data opportunity. Our EduBenchmark archive contains same-item responses from multiple models across teaching and generic tasks, generated under shared harnesses. We first use that archive to screen candidate differences while controlling for item content and generic authorship cues. We then apply paired pedagogy prompts, blinded semantic ratings, human teacher actions, human-gold learner diagnoses, and one small cross-domain affiliation pilot. This archive-first design lets new model calls test specific claims rather than manufacture a personality result from a newly chosen scale.

The findings yield one central distinction. Models have recurring defaults, and a concise prompt can strongly and reliably bias their behavior toward a common teaching style. But a fixed behavioral shift is not personalized education. The question-first prompt improves aggregate action agreement while harming the subset where human teachers chose telling; adaptive-looking prose does not add learner-state information; and models that execute prescribed actions nearly perfectly still struggle to select them from context. The appropriate object of evaluation is therefore neither the model nor the prompt in isolation, but the model–instruction system acting in a learner context.

2 Related Work

Behavior-first psychometrics includes projective stimuli and construct-matched contexts (Wang et al., 2026; Kocielnik et al., 2026). Cross-task work studies assigned-persona consistency across elicitation formats (Reusens et al., 2025). We audit unassigned shared-harness tutor defaults under task, fingerprint, intervention, and external-criterion controls. Our targeted bridge uses public-domain IPIP items (Goldberg et al., 2006) only as one measurement route, not as ground truth for model behavior.

MathDial defines teacher moves and premature solution reveal (Macina et al., 2023); MRBench and MathTutorBench organize open-ended tutoring capabilities (Maurya et al., 2025; Macina et al., 2025). LongTutor separates historical evidence, knowledge-state diagnosis, and adaptive teaching (Li et al., 2026). Pedagogical Alignment and StratL train or optimize a predefined scaffolded policy (Sonkar et al., 2024; Puech et al., 2025); prompt moderation can also harm relative to an unmod-

erated tutor (Steindl et al., 2025). Closest to our context-sensitivity question, Borchers and Shou (2025) find limited adaptivity under learner-context ablations. Tutor-persona steering (Lee et al., 2026) and theory-grounded runtime control (Nkambou et al., 2026) ask how a desired policy can be induced, while PATS adapts strategies to student personality (Rooein et al., 2026). We instead measure recurring deployed-system defaults, audit their construct validity, and then apply the same intervention. Work on tutor-strategy prediction (Ikram et al., 2025) and feedback failure localization (Yasir et al., 2026) further motivates reporting action-specific errors instead of one aggregate tutor score. Strategy-before-generation is itself established: prior work jointly predicts tutor strategies and responses (Wang et al., 2023), while Tutor CoPilot deliberately retains human strategy selection in a live system (Wang et al., 2025). SLOW explicitly separates learner-state reasoning from action selection (Wei et al., 2026). Our routing test is therefore a controlled falsification of prompt-only modularization, not a claim to invent a two-stage tutor.

Answer withholding is already studied under pedagogical jailbreaks and multi-turn adversarial student pressure (Zhao et al., 2026b,a). Real deployment data also show that students commonly seek answers and that context can outweigh intended system design (Kobler et al., 2026). Our learner-action analysis is therefore not a first leakage or steering claim, and policy alignment is not treated as learning benefit.

More generally, IHEval finds sharp degradation when instruction priorities conflict (Zhang et al., 2025). Whether an explicit learner request survives system-level policy clauses is therefore an instruction-hierarchy test, not evidence of empathy, learner understanding, or pedagogical appropriateness.

Attribution and one aggregate score are insufficient: lexical features identify generators across domains, while general helpfulness can reverse between judges (McGovern et al., 2025; Fan et al., 2026). Prompted students are also not assumed to measure learning: recent evaluations find weak linguistic, behavioral, and cognitive fidelity (Scarlatos et al., 2026). We therefore treat objective diagnosis and evidence retrieval as prerequisite competence and reserve learning gain for prospective work.

3 Data and Governance

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The core panel contains six frozen deployed configurations: MiniMax-M3, MiniMax-M2.7, GLM-5.2, DeepSeek-V4-Pro, Doubao-Seed-2.0-Pro, and Qwen3.5-4B. We retain only successful responses available for every model on an item. Six teaching arms yield 31,638 responses; four generic negative-control families yield 57,516. A nine-model MathDial panel adds three configurations for 26,586 responses. LongTutor contributes long learner histories, and MathTutorBench provides paired generic and LearnLM-style pedagogy prompts in standard and hard contexts.

The MathDial source supplies human teacher moves (probing, focus, telling, and generic). LongTutor supplies human-gold knowledge-state diagnosis and reference-scored historical evidence. We pin 84 relevant prediction/summary pairs by SHA-256. Exact provider-response regeneration is impossible: temperature is present for 22/84 runs, while seed and prompt-version metadata are absent from all 84.

Public analysis artifacts store identifiers, hashes, features, scores, and inferred labels rather than source responses. Although the LongTutor paper states CC BY 4.0 for data, we withhold longitudinal learner histories as a privacy-minimizing choice. The external semantic sample was transmitted only after explicit authorization.

4 Methods

4.1 Transparent features and transport

We deterministically count length, structure, formatting, questions, address terms, praise, encouragement, hedges, imperatives, history references, answer reveal, explanation, and diagnosis markers. Features are centered across models within exact item and standardized within benchmark. Multinomial logistic regression predicts model identity while holding out an entire task family. The same protocol on generic tasks is a negative control: strong generic attribution prevents relabeling an authorship fingerprint as pedagogy.

For paired prompts we subtract the generic-arm feature from the pedagogy-arm feature for the same context and model. Bootstrap resampling preserves the six correlated responses within each context. We measure both directional model-by-model effects and cross-model profile dispersion.

4.2 Quality, actions, and learner state

Quality prediction holds out one candidate model at a time. Exact-item effects form the baseline; length and policy features are nested additions. Byte-identical responses scored by two judges provide a judge swap. A separate set of 482 expert positive/negative pairs, each shown in both orders, audits three judges.

Three TF-IDF linear-SVM variants are trained on 18,541 human teacher turns, with five-fold splits grouped by math problem. Generated responses are then compared with the human next move. LongTutor analyses join teaching, diagnosis, and evidence tasks by exact history and model. Held-out-model prediction includes exact-history effects before teaching scores are added.

4.3 Candidate educational tendencies

We chose dimensions from decisions that a tutor must make, not from human trait labels. Four requirements constrained the set: a dimension had to be observable in an open response, educationally interpretable, comparable across heterogeneous tasks, and falsifiable through recurrence or an external criterion. This yields four families.

Assistance allocation separates help directness (revealing an answer, step, or explanation), elicitation (advancing through questions or prompts), and autonomy support (leaving consequential reasoning or choice to the learner). These are related but not opposites: a turn may explain and then elicit, and a question need not preserve learner control. *Learner orientation* separates affective warmth, diagnostic specificity (naming evidence or a misconception), and personalization (adapting content or strategy to this learner); warm language alone need not diagnose or adapt. *Instructional management* covers the information burden imposed in one turn, directive agency, and whether a concrete next action is supplied. *Epistemic expression* separates cautious wording, expressed confidence, revision, and abstention; none is treated as correctness or calibration. Together these families cover how much help is given, how the learner is engaged, how the interaction is advanced, and how knowledge status is communicated.

A candidate counts as recurring only when its model ordering transports across task sources. Reliable annotation, cross-task recurrence, and independent educational validation are separate evidence levels. Correlations among six model config-

Panel or criterion	Models	Shared units	Responses	Primary role
Core teaching	6	5,273 items	31,638	recurring teaching behavior
Generic controls	6	9,586 items	57,516	authorship control
MathDial intervention	9	2,954 item-arms	26,586	steering and dispersion
Blinded semantics	6	358 context-arms	2,148	reliability and transport
LongTutor objective	6	1,000 histories	6,000	diagnosis boundary
Epistemic archive	6	4 sources	–	confidence recurrence
Affiliation pilot	5	12 matched conflicts	280 calls	cross-domain behavior
Human teacher actions	–	18,541 turns	18,541	classifier training
Prospective factorial	5	32 problems	2,560	component addressability
Order replication	5	8 fixed problems	640	sequence robustness

Table 1: Evidence sources and dependence-preserving units. Shared units are not multiplied by the number of models when uncertainty is estimated.

urations are descriptive and are not interpreted as population-level latent factors. Four frozen sources provide confidence and revision measures and five provide unanswerable-abstention rates.

4.4 Blinded semantic panel and frozen hierarchy

The deterministic sample initially contained 360 context-arm batches and 2,160 responses: 40 LongTutor, 80 standard MathDial pairs, 40 hard pairs, and 80 Socratic contexts. Three judges coded eight bipolar 1–5 dimensions: help directness, elicitation, autonomy support, affective warmth, diagnostic specificity, personalization, cognitive load, and epistemic caution. Model identity was hidden and response order independently randomized by judge and batch. Confirmatory scores are judge medians.

One DeepSeek annotation remained unavailable after three rounds of three attempts. Before formal analysis and without inspecting the affected scores, we excluded that generic batch and its prespecified pedagogy counterpart for all judges. The complete-case panel therefore contains 358 batches, 2,148 responses, and 1,074 annotations (0.56% technical attrition).

All eight dimensions remain reported. A frozen evidence hierarchy requires (i) $ICC(3,k) \geq .60$, adequate range, leave-one-judge profile $\rho \geq .70$, and no prompt sign reversal; (ii) nonzero exact-context model variance plus cross-task ICC or median task correlation $\geq .50$; and (iii) an independent quality/action, prompt/action, or BH-corrected human-gold diagnosis criterion. These gates keep reliable measurement, cross-task recurrence, and external behavioral validation as separate claims.

4.5 Prospective component intervention

Before inspecting any generated response or effect, we froze 32 synthetic elementary-math problems

and a complete $2 \times 2 \times 2$ intervention over question-first, answer-reveal, and warm-tone system clauses. Each cell is crossed with learner requests for direct help or exploration and five reachable model routes (2,560 calls). Deterministic detectors score whether the first sentence-like segment is a question, whether the correct answer is revealed, and whether a frozen encouragement lexicon appears. Every cell, model, family, secondary surface outcome, and two- and three-way interaction remains reported.

A component must exceed its registered target-effect threshold (.60 for question and answer; .30 for tone), be positive in every model, and have a target effect at least twice its largest primary cross-effect. Learner-request adaptation has separate .10 thresholds. The parent queue was lexicographic, so, still blind to outcomes, we froze an eight-problem replay with byte-identical prompts, model-specific SHA-256 order, and exact balance of all 16 request-policy cells in every consecutive block. A claim is order-robust only if both panels pass and every model is positive in the matching parent subset and replication; the replay can only downgrade a claim.

After seeing aggregate effects, we froze a downgrade-only detector audit. An outcome-independent hash sample contains 480 responses (320 parent, 160 replay). Three blinded judges label all 48 ten-response batches without model, factor, detector, or effect information. Validation requires majority coverage $\geq .95$, balanced accuracy $\geq .90$ with bootstrap lower bound $\geq .80$, and Cohen’s $\kappa \geq .70$; failure restricts reporting to the literal detector.

4.6 Inference, multiplicity, and freeze timing

The inferential unit follows the claim (Table 1). Cross-task attribution reports an unweighted macro-average over held-out families. Prompt intervals

resample underlying contexts while retaining all model-linked rows; judge calls are first reduced to response medians and never counted as independent educational cases. Quality models report every held-out-model fold, and exact sign tests treat folds, not responses, as replicates. Prompt tests use BH correction over the 48 model $\times$ dimension contrasts within each difficulty; semantic diagnosis uses within-outcome BH correction.

The multi-judge plan was frozen at 311/1,080 annotations, all from the first judge and before any second- or third-judge result. Single-judge directions had been inspected, so this is a prospective multi-judge freeze, not a full preregistration. Manuscript decision thresholds were fixed at 637/1,080, before the third judge and all LongTutor annotations; a later multiplicity amendment also preceded all semantic LongTutor results. Analyzer hashes and the technical- attrition amendment are retained in the audit trail.

4.7 General-personality bridge

Before new generation, a theory-first archive audit mapped Big Five, HEXACO honesty–humility, Dark Triad, Schwartz values, adjacent validity constructs, measurement routes, and stability perturbations to existing tasks. Candidate language profiles had to recur outside tutoring, transport to default tutoring, and survive verbosity and organizational-style controls. The audit could select, but not confirm, one prospective follow-up.

It selected affiliation. We froze five model configurations, twelve matched educational/non-educational conflicts, four conditions (default, irrelevant interface context, high affiliation, and low affiliation), ten IPIP agreeableness items, and twelve forced choices. Three blinded judges rated open responses on five dimensions. Default stability jointly required $ICC(3,k) \geq .70$, cross-domain and perturbation profile $\rho \geq .70$, absolute mean perturbation $\leq .25$, and model $SD \geq .15$. Prompt steerability and convergence of self-report/choice with open behavior were separate frozen claims.

5 Results

5.1 Models differ, but few differences recur

The clearest recurring educational difference is how readily a model gives direct help. Blinded ratings of help directness have high judge reliability ($ICC(3,k)=.905$), recur across teaching tasks (cross-task $ICC=.629$; median task $\rho = .647$), and

agree with an independent behavioral criterion: responses classified as human-like telling score 1.360 points more direct than probing/focus responses. Default model means span roughly 1.65 points, from GLM at the least direct end to MiniMax-M2.7 at the most direct end. This is the strongest evidence in the archive for a recurring educational tendency rather than an impression from one benchmark (Figure 1A).

Expressed confidence provides a second, narrower pattern. Across four independent sources, its model profile is stable ($ICC(3,1)=.852$; median task $\rho = .886$), with Doubao highest (.977) and MiniMax-M3 lowest (.878). This pattern concerns how firmly an answer is expressed, not whether the answer is correct. It also does not collapse revision and abstention: Qwen combines high expressed confidence (.956) with the highest abstention on unanswerable items (.912).

Table 2 turns these results into model-level evaluations without converting them into a quality ranking. Directness is an exact-item centered default score; prompt movement is the mean change on the 1–5 semantic scale across standard and hard contexts. A more direct or more steerable model is not inherently a better tutor.

Some other measurements are repeatable without supporting an equally strong interpretation. Cognitive load recurs across tasks ($ICC=.663$; median $\rho = .677$), while elicitation is reliably rated but less stable across tasks ($ICC=.475$; median $\rho = .468$). Table 3 reports every prespecified dimension, including failures, rather than retaining only favorable ones.

These semantic results are not reducible to response length. Semantic dimensions identify models on held-out teaching tasks at .322, compared with .239 for length and .167 chance. However, generic tasks are even easier to attribute (.372 with combined transparent features versus .242 for teaching policy features), and cross-domain policy geometry is not preserved (Mantel $r = .296$, $p = .307$). Stable authorship cues are therefore a necessary negative control: recognizing a model is not sufficient evidence of an educational personality.

5.2 A shared prompt imposes a common bias, not a common tutor

The same explicit pedagogy prompt causes a large, directionally consistent behavioral shift. All nine models ask more questions and produce shorter

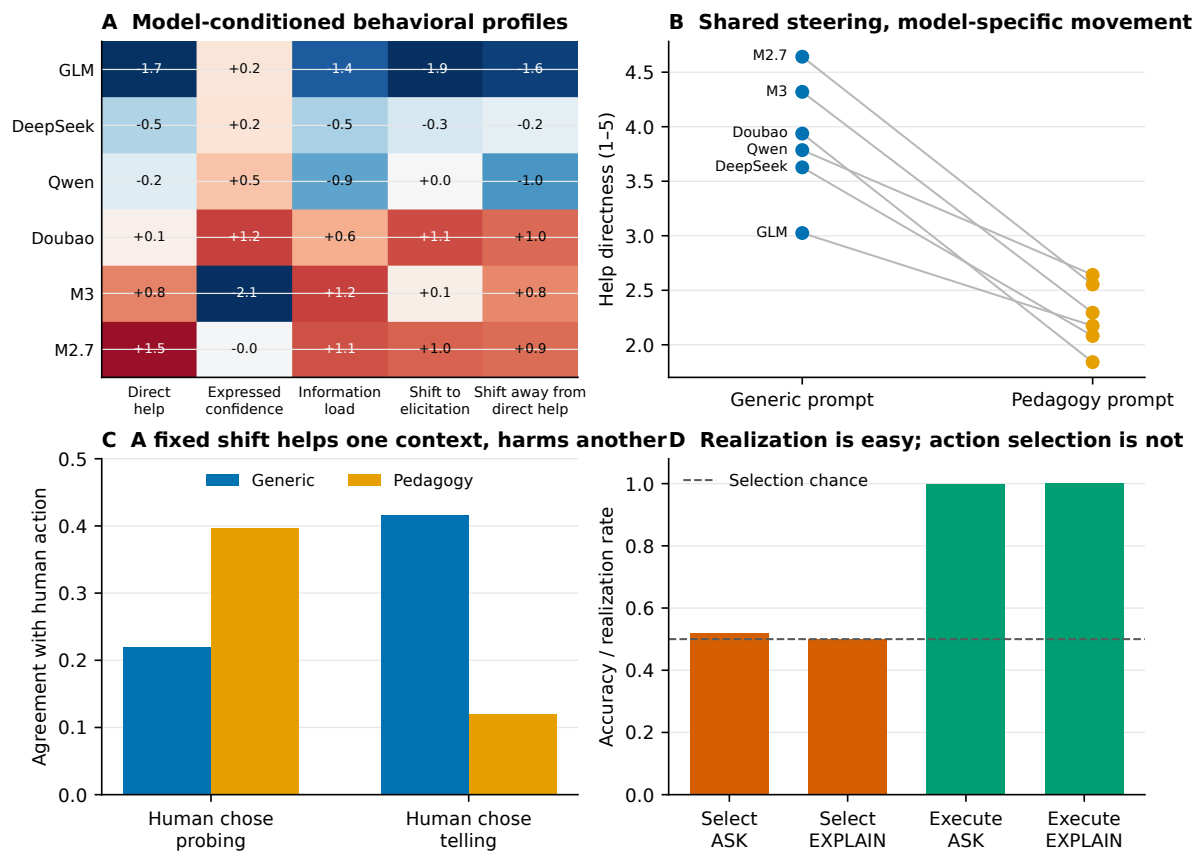


Figure 1: Steering changes a teaching style without providing personalization. (A) Six frozen model configurations have different five-axis behavioral profiles (column-standardized; descriptive, not ability scores). (B) A shared pedagogy prompt moves every model away from direct help, but by different amounts. (C) The fixed shift improves agreement when human teachers probed and harms it when they told. (D) Models can realize prescribed actions but the tested selectors cannot reliably choose them from context.

turns in both standard and hard contexts, cross-model dispersion falls to .763 of its generic-prompt baseline (95% CI [.738, .794]). In blinded semantic ratings, the prompt reduces help directness by 1.709 points in standard and 1.542 in hard contexts, increases elicitation by 2.470 and 2.171, and reduces cognitive load by .759 and .721. The shift in help directness is about as large as the entire default-model range.

This is effective steering: a concise instruction reliably biases the distribution of observable teaching moves. It is not personalization, because every learner context is pushed in the same direction. Nor does it make models interchangeable. Cross-prompt model attribution remains .301 and .297, both above the .167 chance level. Models also differ in prompt responsiveness after accounting for available headroom, and some pairwise model rankings reverse after prompting. The evidence therefore supports neither fixed personality nor complete prompt determination: models have distinguish-

able defaults, a shared prompt moves all of them, and the amount and sometimes direction of residual movement is model-specific. Prompt steering should therefore be evaluated for each model configuration rather than treated as a portable, model-agnostic teaching layer.

5.3 Several intuitive personalities do not survive

The archive contains many plausible-looking differences that fail once recurrence, discriminant validity, or behavioral transfer is required. The most instructive case is next-step actionability. The pedagogy prompt increases it by 1.047 points in standard and .880 in hard contexts for every model, so it is highly steerable. Yet its cross-task ICC is $-.111$ and its median task correlation is $-.771$: the model ordering reverses rather than recurs. Steerability is therefore not evidence of a stable default.

Relational communion is reliably measured (ICC=.945), but correlates .805 with surface

Configuration	Default directness	Expressed confidence	Prompt Δ directness	Evidence-bounded reading
MiniMax-M2.7	.803	.941	−2.089	most direct default; large prompt movement
MiniMax-M3	.408	.878	−2.026	direct default; lowest expressed confidence; large movement
Doubao-Seed-2.0-Pro	.071	.977	−2.096	near-mean directness; highest confidence; large movement
Qwen3.5-4B	−.113	.956	−1.144	slightly less direct; high confidence; smaller movement
DeepSeek-V4-Pro	−.324	.948	−1.546	less direct default; intermediate movement
GLM-5.2	−.845	.948	−.851	least direct default; smallest additional movement

Table 2: Evidence-bounded profiles for the six-model core panel. Values describe frozen deployed configurations under the sampled tasks and prompts; they are neither family-level traits nor pedagogical-quality scores.

Dimension	ICC(3,k)	Task ICC	Median task ρ	Reversals	Highest retained tier
Help directness	.905	.629	.647	0	recurring behavior + criterion
Elicitation	.963	.475	.468	0	reliable measurement only
Autonomy support	.819	.610	.734	1	none
Affective warmth	.937	.479	.714	3	none
Diagnostic specificity	.924	.203	.371	1	none
Personalization	.832	.340	.257	1	none
Cognitive load	.855	.663	.677	0	recurring behavior
Epistemic caution	.580	.035	.129	0	none

Table 3: Every prespecified semantic dimension remains visible. Reliable measurement and cross-task recurrence are separate from external behavioral validation.

warmth and shows no model variance in the Socratic panel. Instructional agency remains exploratory because its frozen pilot reliability (.657) misses the .70 threshold. Personalization and diagnostic specificity also fail cross-task recurrence, while learner requests do not reliably change the question/answer behaviors they request. These results contradict the intuitive idea that warm or adaptive-sounding language demonstrates learner-responsive teaching.

Attempts to translate generic human traits fare no better. Organizational style recurs outside tutoring (median $\rho = .771$), but correlates negatively with IFEval accuracy ($\rho = -.886$, exact $p = .035$), so it cannot be treated as conscientiousness. Cautious wording does not establish honesty–humility, and a proposed normative-permissiveness axis reverses across its two candidate tasks ($\rho = -.486$). The honest conclusion is not that these models lack all such traits, but that the present archive does not identify them.

A targeted five-model pilot finds affiliative open behavior that transports across educational and non-educational conflicts ($\rho = .900$) and survives one irrelevant perturbation ($\rho = .718$). Yet IPIP self-report correlates $-.200$ with open behavior, forced choice is at ceiling, and open affiliation correlates $.900$ with warmth. This is limited behavioral evidence, not Big Five agreeableness.

5.4 Steering a style is not personalizing instruction

Prompting all models toward questions improves aggregate agreement with human teacher actions: all 54 classifier–difficulty–model contrasts rise by .091–.114 in standard and .145–.180 in hard contexts. The aggregate gain, however, comes mainly from contexts dominated by probing. In the 48 standard contexts where human teachers chose telling, generic-prompt agreement is .394–.442 and pedagogy-prompt agreement falls to .111–.125. The inferred telling rate drops by .234–.280. A recognizable “Socratic” style can thus be consistently induced and consistently wrong for the local teaching decision (Figure 1C). The prompt changes what the tutor tends to do; it does not infer what this learner needs. Aggregate improvement and contextual adaptation are different claims.

The learner-state results reveal a second separation. Across 6,000 model–history observations, teaching quality relates only weakly to diagnosis (mean within-model Spearman $\rho = .173$) and not to evidence accuracy ($-.010$). With the exact history controlled, item-only diagnosis reaches AUC $\rho = .822$; adding four teaching dimensions yields .816, and their mean yields .817. Fluent, adaptive-sounding teaching does not supply the missing diagnosis.

The prospective factorial explains part of this gap. Explicit system clauses strongly control question-first and answer-reveal behavior and replicate under a different call order. In contrast, learner

requests for direct help versus exploration change answer reveal by only .073 in the parent panel and .066 in replication, below the frozen .10 gate; their effect on question-first is small and opposite-signed. Models follow the system-level teaching instruction much more reliably than the learner's expressed need.

All three operational endpoints pass every registered parent gate and the downgrade-only order replication. Effects are positive for all five models in the full panel, matching parent subset, and replication. By family, question effects span .750–.787 in the parent and .750–.863 in replication; answer effects span .878–.928 and .887–.975; encouragement-marker effects span .525–.659 and .487–.600. Thus no one model or problem family supplies the aggregate result.

A blinded detector audit validates question-first and answer reveal against human labels (balanced accuracy 1.000 and .996). The encouragement detector does not pass its semantic validation gates (balanced accuracy=.818, $\kappa = .631$), so the third row supports only use of the frozen lexicon, not a claim that the response is genuinely warm or supportive.

Execution is not the main unsolved problem. When externally assigned ASK or EXPLAIN, models realize the action at .998 and 1.000. Two selectors reach only .519 and .500 accuracy, and composing selection with realization performs worse than a single-pass tutor (−.058 and −.077). Stable educational style mainly describes what a model tends to do; it does not show that the model knows when to do it.

6 Discussion

Recurring behavior remains conditional. Help directness and expressed confidence recur, but most plausible dimensions do not, and even the recurring differences are substantially displaced by a shared instruction. Residual model identity, unequal responsiveness, and rank reversals show that neither a default score nor a prompt effect describes the tutor on its own. What recurs is a model-conditioned tendency expressed under a task and an instruction. This conditional account is also educationally important: a stable preference for telling or eliciting describes how support is allocated, not whether that allocation fits a particular learner. Generic authorship signatures may likewise recur without transporting the educational construct, so model

attribution alone is insufficient.

A fixed shift cannot substitute for adaptation.

The pedagogy prompt reliably changes how tutors act, but it does not establish that the induced action is appropriate. Its average gain comes from probing contexts, while agreement drops sharply where human teachers told. Teaching presentation adds no learner-state signal, and near-perfect action realization coexists with near-chance action selection. These results locate the limitation more precisely than a generic claim that prompting “does not work”: prompting is effective for moving a behavioral distribution, but personalization requires evidence about the learner and a context-conditioned choice among teaching actions. Learning outcomes remain a further, untested requirement.

The tutor system is the relevant unit of evaluation.

Our intervention varies prompts and system-level clauses inside a fixed generation setup, not a full harness. Memory, retrieval, learner modeling, action routing, tools, or post-generation checks could address limitations of a fixed prompt. Yet the present results caution against treating the base model as a neutral, interchangeable executor: the same instruction produces different movement across models, so changing the model can change the resulting tutor even when the surrounding logic is held constant. End-to-end evaluation should therefore pair each intended model with its instructional setup and report recurring defaults, induced change, model-specific responsiveness, action-selection accuracy, and context-specific outcomes. The same discipline applies to construct claims: inducible actionability is not stable, communion largely redescribes warmth, organization does not validate conscientiousness, and agreeable self-report does not predict open behavior. Consistency, steerability, adaptation, and efficacy remain distinct targets.

7 Conclusion

AI tutors in this finite panel differ in educational behavior, and a small subset of those differences recurs, but the result is not a single fixed personality and must not be equated with human traits or pedagogical quality. A shared prompt can strongly steer every model without making models equivalent. The central boundary is educational: steering a tutor toward questions, restraint, or warmth does not personalize instruction. The same fixed shift

can help probing contexts and harm telling contexts, while models that execute a prescribed action nearly perfectly still struggle to select it from the learner context. AI-tutor evaluation should therefore target the model–instruction–context system and keep behavioral recurrence, steering, action selection, and learning outcomes separate.

Limitations

The core panel contains six systems; response count does not create model replicates, so population and family claims are excluded. Most quality outcomes are LLM-judged, with judge swaps, expert-pair calibration, human action labels, and human-gold diagnosis as checks. Human action agreement is not learning gain; efficacy requires prospective learners and objective pre/post tests.

The affiliation pilot contains five configurations, twelve synthetic conflicts, and one irrelevant perturbation. Exact profile tests are underpowered, generator/judge families overlap, and the primary score is not discriminant from surface warmth. Broader perturbations, behavioral games, and an independent larger model panel are required before any general-personality claim.

The prospective factorial uses synthetic elementary-math contexts and surface detectors. A post-result blind audit validates question-first and correct-answer reveal, but the encouragement lexicon fails semantic warmth validation and is reported literally. Its five endpoints are provider snapshots, and the eight-problem replay occurred later; provider-version drift and repeat-prompt caching cannot be separated from request-order robustness. Component effects are not learning outcomes, and reproduced interactions rule out treating the clauses as independent latent traits.

Provider system prompts, alias updates, and serving backends are uncontrolled. Generation cannot be exactly repeated because most settings and all seeds and prompt versions were not retained. Findings concern frozen deployed configurations, not immutable weights. The semantic complete-case rule was fixed after a provider failure but before formal analysis; its 0.56% paired technical attrition is fully reported.

Ethics Statement

The study analyzes existing model responses and educational histories. Public artifacts use hashes and derived measurements rather than raw learner

histories or source responses. External semantic coding used only explicitly authorized endpoints and a frozen payload. Model-conditioned differences must not be read as human traits, demographic characteristics, or evidence that one system improves learning. Deployment should test learner outcomes and provide educator oversight, especially where a uniform Socratic policy may withhold needed direct help.

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